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# Light Inhibits Lens-Induced Myopia through an Intensity-Dependent Dopaminergic Mechanism

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***Purpose:*** Bright light exposure has been postulated to underlie the ability of time spent outdoors to prevent the development of myopia in humans. In support of this, bright light inhibits the development of form-deprivation myopia (FDM) in all species studied. While lens-induced myopia (LIM) is also inhibited by bright light in most species, it remains unclear whether this is brought about in an intensity-dependent manner and whether dopamine (DA) plays the same critical role in this paradigm as is seen in FDM.

***Design:*** An experimental study.

***Subjects:*** White Leghorn chickens (*Gallus gallus*).

***Methods:*** To examine the effect of light on LIM, chicks fit with lenses of −10 diopters were exposed to 500, 20 000, or 40 000 lux for 14 days (n = 6 per group). To assess the role of DA, its levels were measured 30 minutes after light exposure in previously dark-adapted animals over 6 light intensities (between dark and 40 000 lux). In a separate experiment, a D1-like (SCH-23390) or D2-like (spiperone) receptor antagonist was administered (once daily) to chicks wearing negative lenses under 40 000 lux (n = 5 to 6 per group) for a period of 5 days.

***Main Outcome Measures:*** Refraction (infrared photoretinoscopy), axial length (A-scan ultrasonography), and DA levels (liquid chromatography-tandem mass spectrometry).

***Results:*** Bright light inhibited LIM in an intensity-dependent manner ($P < 0.05$) but did not prevent full compensation. The protection afforded by bright light was significantly reduced by administration of spiperone (D2-like, $P < 0.05$), but not SCH-23390 (D1-like, $P = 0.77$). Retinal DA levels showed an intensity-dependent increase ($P < 0.05$).

***Conclusions:*** As previously observed for FDM, bright light can inhibit the development of LIM in an intensity-dependent manner. This protection occurs, at least in part, via a DA-dependent mechanism. However, bright light's inability to prevent compensation to negative lenses is indicative of mechanistic differences between the 2 experimental models of myopia. These differences are most likely linked to the presence of a defocus-driven end point for growth in LIM that is not present in FDM.

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*Supplemental material available at www.ophthalmologyscience.org.*

The prevalence of "school myopia," a visual disorder caused (predominantly) by excessive elongation of the eye during development, has reached epidemic proportions in many parts of the world (most notably in urban areas of East Asia) (for review, see[1]). Epidemiological studies have shown a strong link between reduced time outdoors and the increased prevalence of this disorder.[2–5] This epidemiological observation has since been supported by several randomized clinical trials that have demonstrated a reduction in myopia onset after increasing the time children spend outdoors during a school day.[6,7] Rose et al[3] postulated that such protection might involve the greater light intensities experienced outdoors and, with it, an associated increase in the release of dopamine (DA). This hypothesis was based on known links between brighter and flickering light on DA release and the known ability of DA agonists to block axial elongation in animal studies.[8–12]

This hypothesis has been supported by experimental studies across a range of species. Specifically, in chickens,[13,14] tree shrews,[15] mice,[16] guinea pigs,[17] and rhesus monkeys,[18] exposure to bright light (>2000 lux) has been shown to inhibit the development of form-deprivation myopia (FDM). Bright light has also been shown to inhibit the other major form of experimental myopia (lens-induced myopia [LIM]) in chicks,[19] guinea pigs,[20] and tree shrews.[15] In rhesus monkeys, while bright light blocks the development of FDM, it has been reported to be ineffective at inhibiting the development of LIM (although a small trend is observed).[21]

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While the results on rhesus monkeys have been used to argue that outdoor light exposures may not be relevant to the prevention of human myopia, there is evidence from human studies that would support a link. Read et al[22] observed that greater daily light exposure was associated with less axial growth in children. In school-based randomized trials, the protective effects of time outdoors correlate roughly with the total duration and intensity of light exposure.[23] Further, myopia progression is slower in summer than in winter, with a corresponding difference in daily light exposure.[24,25] However, it is not possible to infer a direct causal role for light in the antimyopic effects observed in these studies. Light exposure could simply be a marker of time outdoors rather than being the underlying driver. However, in direct support for luminance levels playing a role in modulating ocular growth rates, a 12-month study by Hua et al[26] reported that increasing classroom illumination by six fold reduced the onset and progression of myopia in children. This study, however, requires replication.

In FDM, work in chicks has shown that experimental myopia can be inhibited in an intensity-dependent manner.[14] Further, protection is driven, at least in part, by the dopaminergic system, as the antimyopic effects of bright light are lost when the DA D2-like receptor antagonist spiperone is administered.[19] What remains unclear is whether LIM is also inhibited in an intensity-dependent manner and whether DA plays the same critical role in this paradigm, 2 questions that will be addressed by this study.

## Methods

### Animals and Housing

One hundred eighty-six chickens (one-day-old male White Leghorns) were obtained from Barter & Sons Hatchery or from Aviagen Hatchery. Chicks were housed in temperature-controlled rooms (with unlimited access to food and water) and were given 4 days to adjust to their surroundings before the commencement of experiments. Lights were on a 12:12 hour light:dark cycle (lights on at 9 AM and off at 9 PM). Authorization to conduct experiments using animals was approved by the University of Canberra Animal Ethics Committee under the ACT Animal Welfare Act 1992 (project number: CEAE 2098) and conformed to the Association for Research in Vision and Ophthalmology Statement for the Use of Animals in Ophthalmic and Vision Research.

### Myopia Induction

As previously described,[27] myopia was induced in chickens by placing either a −10 diopters (D) powered lens or a translucent diffuser over the left (treated) eye. In all experiments, the left eye served as the experimental eye, while the right eye remained untreated, serving as an internal contralateral control eye. Age-matched untreated animals were also included within each experiment. For all experiments, after baseline ultrasound and refractive measurements on the day before the commencement of treatment, Velcro mounts were fit around the left eye with Loctite super glue (Henkel). On the following day, the negative lenses or diffusers, fit to matching Velcro rings, were placed onto the mounts.

The 2 forms of blur (diffuse [FDM] vs. defocus [LIM]) used in this study produce different levels of image degradation and will thus have dissimilar spatial frequency spectra. The diffusers degrade contrast sensitivity at all spatial frequencies (but with greater effect at higher frequencies), providing (for the most part) only light perception (similar to a Bangerter foil with a strength of between 0.1 and 0.0). The diffusers also produce a loss in light transmittance (∼15%) due to backscatter, absorption by the material, as well as forward scatter that falls outside of the pupil. In contrast, the negative lenses affect contrast sensitivity at mid and high spatial frequencies (with little effect at lower spatial frequencies) and have minimal effect on light transmittance (∼3%). Importantly, unlike diffuse blur, the defocus-induced alterations in visual perception can be compensated for by accommodation (the accommodative power of the chicken is approximately 20 D[28]).

### Measurements of Myopia Induction

As previously described,[27] refraction and axial length were measured by automated infrared (IR) photoretinoscopy (system provided by Professor Frank Schaeffel, University of Tuebingen) and A-scan ultrasonography (Biometer AL-100, Tomey Corporation), respectively.

Ultrasonography measurements were performed on chicks anesthetized under light isoflurane (5% in 1 L of medical-grade oxygen per minute, Veterinary Companies of Australia) using a vaporizer gas system (Stinger Research Anesthetic Gas Machine [2848], Advanced Anaesthesia Specialists). Ultrasound measurements were performed using an immersion attachment probe filled with medical grade ultrasound gel. For each eye, ultrasonography measures represent the average of 3 scans, with each scan comprised of the average of 10 consecutive scans.

The refractive values reported represent the mean spherical equivalent of 10 measurements per eye. Refractive measurements were carried out on conscious, noncycloplegic chickens in darkened rooms (<5 lux).

### Experiment 1: Studying the Effects of Bright Light on Negative-Lens Compensation

For high-light treatment, animals were exposed to 1 of 3 lighting levels (500 lux, 20 000 lux, or 40 000 lux) for a period of 6 hours each day (11 AM till 5 PM) commencing 2 hours after normal light onset (treatment groups and numbers of animals are detailed in Table 1, while the experimental protocol is detailed in Fig S1, available at www.ophthalmologyscience.org). For the remainder of the light phase (9 AM till 11 AM and 5 PM till 9 PM), animals were kept under a "normal" room illuminance of 500 lux. For animals in the untreated control group, as well as the 500 lux myopia group, these chicks were housed under 500 lux room illuminance for the entire 12 hour light phase. The experiment continued until all 3 lighting groups had compensated for the −10 D lens (14 days).

### Experiment 2: Assessing the Reproducibility of Bright-Light's Protection

To evaluate the reproducibility of the antimyopic effects of bright light exposure (40 000 lux), the rate of compensation to negative lenses over 5 days was compared across 4 different batches of animals. This included the chickens from experiment 1, which investigated bright light's impact on lens compensation (batch 1), and 3 additional batches (Table 1, Fig S1). In each experiment, animals fit with −10 D lenses were exposed to 40 000 lux for 6 hours each day over a period of 5 days. These animals were compared with untreated controls and animals wearing −10 D

Table 1. Treatment Allocations of Animals across Experiments

| Condition | Number of Animals |
|---|---|
| Experiment 1: The effects of bright light on negative-lens compensation | |
| Untreated | 6 |
| LIM (500 lux) | 6 |
| LIM (20 000 lux) | 6 |
| LIM (40 000 lux) | 6 |
| Experiment 2: The reproducibility of bright light's protection | |
| Untreated | 5 per batch (3 batches)* |
| LIM (500 lux) | 5 per batch (3 batches)* |
| LIM (40 000 lux) | 5 per batch (3 batches)* |
| Experiment 3: The role of dopamine in bright light's antimyopic effects | |
| Untreated | 4 |
| LIM (500 lux) | 9 |
| LIM (40 000 lux) | 6 |
| LIM + Vehicle (40 000 lux) | 6 |
| LIM + D2-like dopamine antagonist spiperone (40 000 lux) | 6 |
| LIM + D1-like dopamine antagonist SCH-23390 (40 000 lux) | 5 |
| Experiment 4: The effects of bright light on dopamine levels | |
| 0 lux | 5 |
| 50 lux | 5 |
| 500 lux | 5 |
| 10 000 lux | 5 |
| 20 000 lux | 5 |
| 40 000 lux | 5 |
| Experiment 4: The effects of AP4 on electroretinogram recordings | |
| Untreated | 5 |
| Metabotropic glutamate agonist AP4 | 5 |
| Experiment 4: The effects of vehicle solution on electroretinogram recordings | |
| Untreated | 5 |
| Vehicle | 5 |
| Experiment 4: The effects of bright light and AP4 on dopamine levels | |
| Untreated (500 lux) | 5 |
| FDM (500 lux) | 5 |
| LIM (500 lux) | 5 |
| FDM (40 000 lux) | 5 |
| LIM (40 000 lux) | 5 |
| FDM + metabotropic glutamate agonist AP4 (40 000 lux) | 5 |
| LIM + metabotropic glutamate agonist AP4 (40 000 lux) | 5 |

AP4 = L-(+)-2-amino-4-phosphonobutyric acid; FDM = form-deprivation myopia; LIM = lens-induced myopia.
Vehicle: 0.1% ascorbic acid in 1x phosphate buffered saline.
*Experiment 2 involved 3 repeated batches of animals (n = 5 per batch), plus data from experiment 1.

lenses under 500 lux for the entire light cycle. Refraction was measured daily, with axial length measured at baseline and completion of the experiment.

## Experiment 3: Examining How DA Receptor Blockade Attenuates the Protective Effects of Bright Light

To assess the role of DA in the antimyopic effects of bright light, animals fit with −10 D lenses and exposed to 40 000 lux were intravitreally injected immediately before bright light exposure with spiperone (D2-like DA receptor antagonist; S7395, Sigma-Aldrich), SCH-23390 (D1-like DA receptor antagonist; D054, Sigma-Aldrich), a placebo solution (0.1% ascorbic acid dissolved in 1× phosphate buffered saline), or no injection (Table 1, Fig S1). As with the previous experiments, animals were exposed to bright light for a period of 6 hours each day for the length of the experiment (5 days). Intravitreal injections (using a 30-gauge needle (Terumo) fit to a Hamilton syringe) were performed under light isoflurane anesthesia (following the anesthesia method outlined earlier). Ten microliters of spiperone and SCH-23390 were injected daily at a concentration of 0.5 millimole each. Drugs were dissolved fresh each day in 0.1% ascorbic acid (pH 7.4) as previously described.[19]

## Experiment 4: Measures of Ocular DA Levels and the Role of ON-Bipolar Cell Input

To assess the correlation between light intensity and DA levels (DA release, uptake, and breakdown), the DA metabolite 3,4-dihydroxyphenylacetic acid (DOPAC) was measured in vitreal samples 30 minutes after exposure to 1 of the following 6 light intensities (dark [∼0 lux], 50 lux, 500 lux, 10 000 lux, 20 000 lux, and 40 000 lux; Table 1) in otherwise untreated animals (no lenses or diffusers). Before light exposure, animals were dark-adapted overnight for a period of 12 hours.

To remove the vitreal body, chickens were heavily anesthetized using isoflurane and sacrificed by decapitation. Each eye was rapidly removed and hemisected equatorially to float both the anterior and posterior eye cup in 1× phosphate buffered saline

(pH 7.4; Sigma Aldrich). This facilitated collection of the vitreous, which was then snap frozen and stored at −80°C until analysis. As previously described, DOPAC was measured using liquid chromatography tandem mass spectrometry[29] (see Appendix and Table S2, available at www.ophthalmologyscience.org for full liquid chromatography tandem mass spectrometry methods) using an Agilent 1260 Infinity HPLC interfaced with an Agilent 6410 quadrupole mass spectrometer.

To assess what drives the light-induced change in extracellular DA levels, a subsequent experiment studied the effects of L-(+)-2-amino-4-phosphonobutyric acid (AP4; Sigma A7929) on extracellular DA levels during the development of FDM and LIM. To reduce input from classical photoreceptor pathways, AP4, which acts as a metabotropic glutamate receptor agonist, blocks the action of rod-bipolar and cone ON-bipolar cells within the retina. Before studying its effects on extracellular DA levels, the effects of AP4 on retinal function were validated through electroretinogram (ERG) measurements. After overnight dark adaptation, chicks were either left untreated or given a 10 µL intravitreal injection of a 10 mM AP4 solution 30 minutes before ERG recordings (n = 5 per group, Table 1). As previously described, ERG measurements were undertaken[30,31] (see Appendix for full ERG methods).

After the validation of AP4's effects on retinal function, chicks were fit with diffusers or −10 D lenses and exposed to either 40 000 lux or 500 lux. For the 40 000 lux groups, a subsection of animals was given a 10 µL intravitreal injection of a 10 mM AP4 solution (under dim red light in dark-adapted animals) immediately before light onset (Table 1, Fig S1). After 2 hours of light exposure, animals were sacrificed to allow the measurement of vitreal DOPAC levels following the methods outlined earlier.

## Lighting System Parameters

As previously described,[14] for all bright light groups (10 000 lux, 20 000 lux, and 40 000 lux), illumination of the cage was produced by a custom-made system composed of 10 banks of light-emitting diode (LED) lamps (Electronics Department, Research School of Biology, Australian National University). Illumination output was adjustable on a linear scale (between 0 and 40 000 lux). As used in our previous FDM study,[14] each LED bank was composed of an equal mix of cool (400−650 nm, peaking at 450 nm) and warm (430−700 nm, peaking at 630 nm) LED modules that did not emit in either the IR or ultraviolet (UV) spectrum. The mixture of cool and warm LED light banks was used to broaden the spectral makeup of the system. For 500 lux, cages were illuminated by the room's normal fluorescent lighting system (Phillips TLD36W840ALTO [correlated color temperature 4000K]). During the 12 hour light cycle, when not exposed to bright LED light, chicks were kept under this fluorescent 500 lux system. Illumination levels were uniform across the cage area (1.6 × 0.8 m), as measured by an IL-1700 Research Radiometer (International Light, Inc). All luminance intensities reported in this study were measured at the level of the cage floor with the light meter probe pointing upward.

## Spectral Distribution

The floor-level irradiance (at wavelengths between 200 and 1000 nm) was quantified in µW/cm$^2$ using a spectroradiometer fit with a cosine diffuser (USB2000+XR1-ES; Ocean Optics). Six light sources were analyzed:

1. Fluorescent light (500 lux): normal room lighting in the animal facilities.
2. LED "cold white" (∼5000 lux): a component of the bright "white" light LED rig.
3. LED "warm white" (∼5000 lux): a component of the bright "white" light LED rig.
4. LED "cold and warm white combined" (10 000 lux): bright "white" light LED rig.
5. Halogen "white light" (10 000 lux): one of the earlier systems used to generate bright light.[13,19]
6. Sunlight (∼90 000 lux): as measured in the early afternoon, in late summer (February), Canberra, Australia.

The spectral distribution of each light source is shown in Figure 2 and Dataset S1 (available at www.ophthalmologyscience.org). Each of the artificial light sources did not emit in the UV spectrum, with only the halogen emitting in the IR spectrum. While sunlight emitted in both the UV and IR spectrums, the highest irradiances were observed in the visual spectrum, with a slightly higher irradiance in the blue end of the visual spectrum. Compared with sunlight, the emittance of the artificial light sources tested was less evenly distributed across the spectrum. Of these, the most spectrally defined emittance was observed under fluorescent lighting, with 3 major narrow-band peaks. A more even spectral distribution was observed for LED lighting. However, 2 distinct peaks remained, 1 peaking in blue and the other in red/green. A wider band emittance was observed under halogen illuminance that was heavily biased toward the red end of the spectrum. All light sources tested emitted wavelengths capable of stimulating each of the currently known photopigments of the chick retina (spectral sensitivity for each opsin type is given as an overlay in Figure 2, labeled a-i). The spectral sensitivity of each photoreceptor type was calculated using Govardovskii nomograms[32] for a vitamin A1-based opsin of approximate λ max as follows: rod − 506 nm, violet-sensitive cone − 419 nm, short-wavelength-sensitive cone − 455 nm, middle-wavelength-sensitive cone − 508 nm, long-wavelength-sensitive cone − 570 nm, double cone − 565 nm, OPN3 − 470 nm, OPN4 − 480 nm, and OPN5 − 417 nm.[33–37] With regard to the experiments run in this study, it is worth noting that although total photons were reduced by roughly 3% and 15% for lenses and diffusers, respectively, the spectral distribution of light was unaffected by either device.

## Total Effective Photons

While each light source tested is classified as "white light," their spectral distributions would result in quite different activation profiles of the chicken photoreceptors. Therefore, the total effective photons available to each photoreceptor were calculated and compared between the different lighting sources to demonstrate their effects on the chick's retinal experience (Dataset S1). Figure 3 shows the total effective photons for each lighting source (fluorescent, LED, halogen, and sunlight), as calculated by converting total irradiance to photons/cm$^2$/second (as described by Pérez-Fernández et al[38]). Total effective photons for each of the 4 single cones of the chick retina (violet-sensitive, short-wavelength-sensitive, middle-wavelength-sensitive, and long-wavelength-sensitive), as well as rods, were calculated as described by Wilby et al[39] (Dataset S1). This analysis took into account photoreceptor spectral sensitivity, ocular media transmission (as described by Lind and Kelber[40]), photoreceptor transmission (as described by Wilby et al[39]), and the multiplication factor of the eye's aperture (pupil area:retinal area ratio [based on measurements in our laboratory]). The total effective photons receivable by the double cones of the chick retina,[41] as well as the nonvisual opsins (OPN3, OPN4, and OPN5), were not calculated, as their cellular light transmission is not well characterized in the chick.

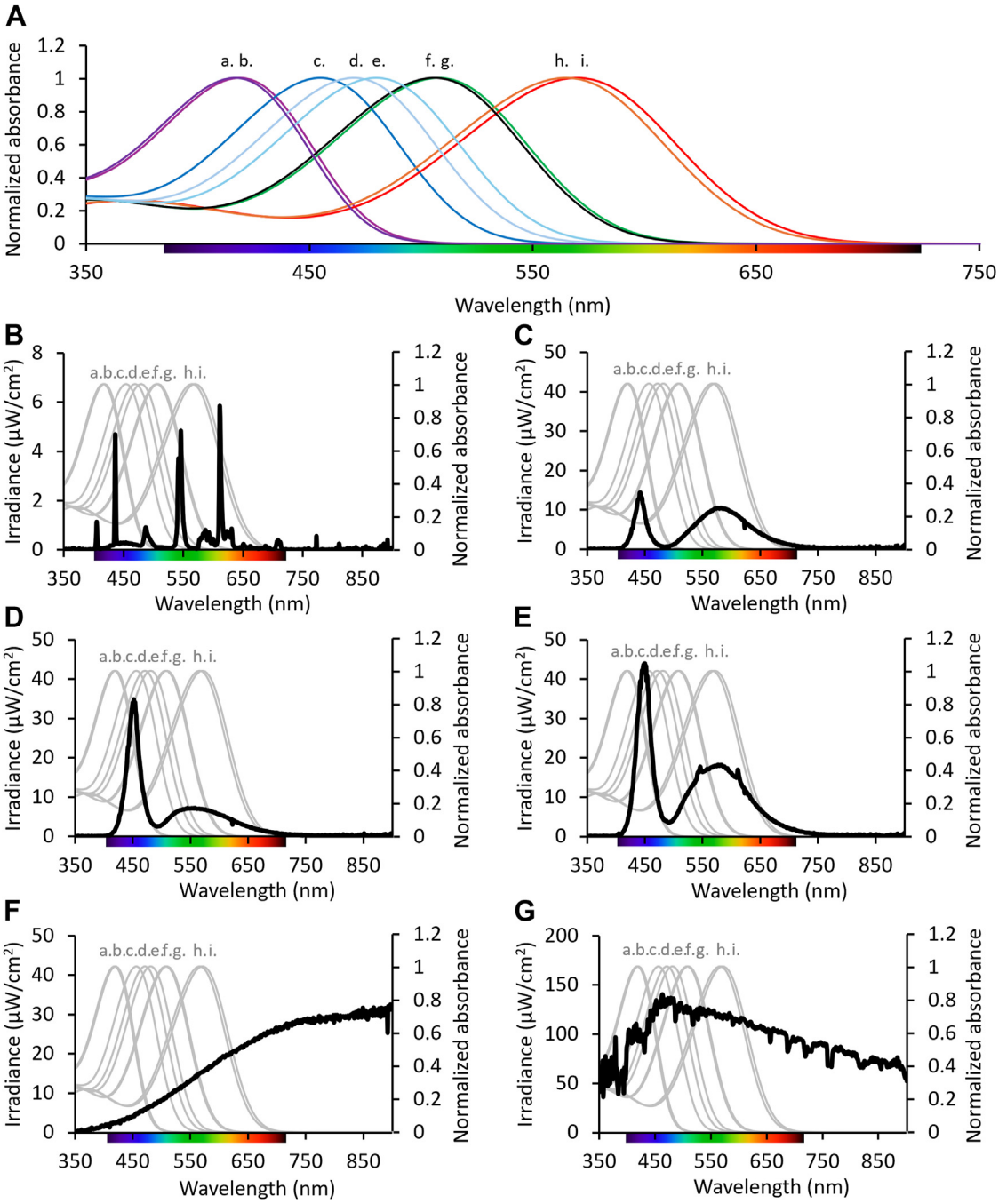



**Figure 2.** Spectral distribution of light sources used in this study. (**A**) Absorbance curves for chick opsins are displayed following calculations as described by Govardovskii et al,[32] using chick opsin λ max values obtained from Hart.[33] The irradiance (black lines) at wavelengths between 350 and 900 nm is presented for (**B**) 500 lux fluorescent lighting, (**C**) ~5000 lux "warm white" LED lighting, (**D**) ~5000 lux "cool white" lighting using LEDs, (**E**) 10 000 lux combined "cool white" and "warm white" lighting using LEDs, (**F**) 10 000 lux halogen lighting (as used in initial studies examining the role of light in experimental myopia[13,19]), and (**G**) ~90 000 lux sunlight (as measured in the early afternoon during summer [Canberra, Australia]). The visual spectrum is displayed in color bars under each panel. Normalized absorbance ranges of chick opsins are depicted in gray underlying each frequency distribution plot. a = OPN5 opsin, b = violet-sensitive cone opsin, c = short-wavelength-sensitive cone opsin, d = OPN3 opsin, e = OPN4 opsin, f = rhodopsin, g = medium-wavelength-sensitive cone opsin, h = double cone opsin, i = long-wavelength cone opsin; LED = light-emitting diode.

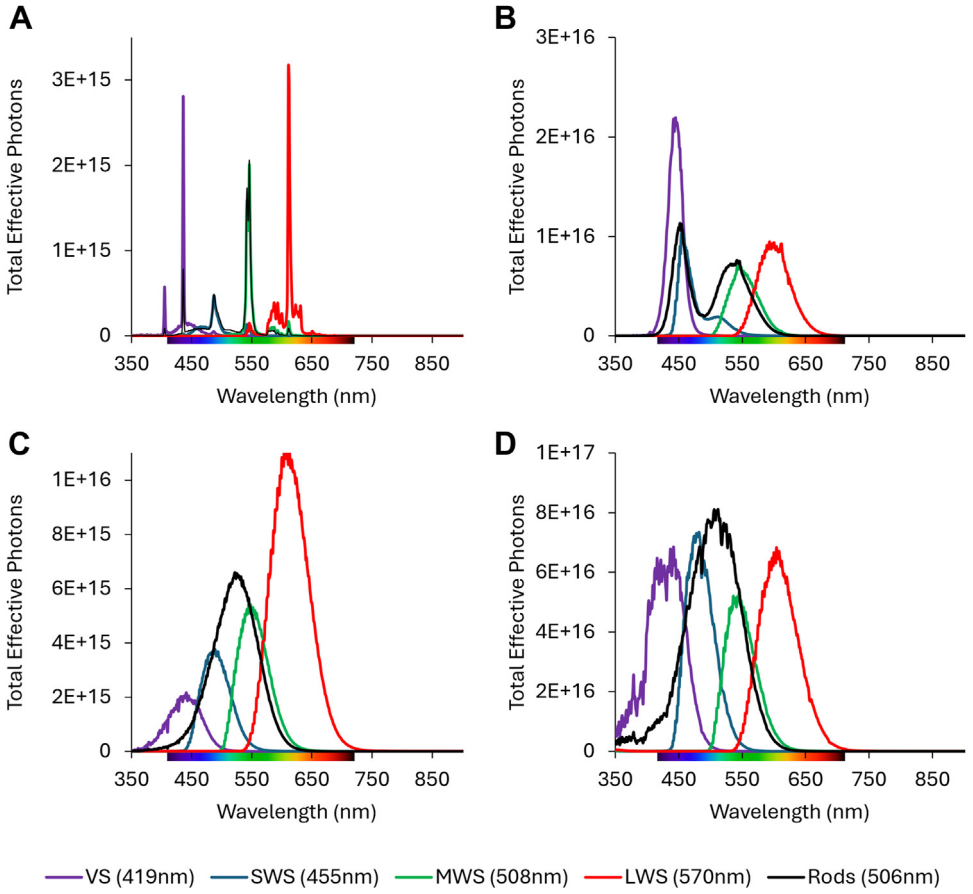



**Figure 3.** Comparison of total effective photons for white light sources. The total effective photons for single cones and rods were calculated between 350 and 900 nm for (**A**) 500 lux fluorescent lighting (note, the peaks depicting total effective photons for rods and MWS cones overlap), (**B**) 10 000 lux combined "cool white" and "warm white" lighting using LEDs, (**C**) 10 000 lux halogen lighting (as used in initial studies examining the role of light in experimental myopia[13,19]), and (**D**) ~90 000 lux sunlight (as measured in the early afternoon during summer [Canberra, Australia]). The visual spectrum is displayed in color bars under each panel. Due to limited data on rod photoreceptor transmittance in the chicken, rod total photons are calculated using only ocular transmittance, relative opsin sensitivity, and aperture corrections. The peak sensitivity (λ max) is displayed in brackets after each opsin at the bottom of the figure. LED = light-emitting diode; LWS = long-wavelength cones; MWS = medium-wavelength-sensitive cones; SWS = short-wavelength-sensitive cones; VS = violet-sensitive cones.

Table 3. Comparison of Summed Total Effective Photons for White Light Sources

| Light Source | VS (419 nm) | SWS (455 nm) | MWS (508 nm) | LWS (570 nm) | Rods (506 nm) |
|---|---|---|---|---|---|
| Fluorescent (500 lux) | | | | | |
| Total effective photons | $1.3 \times 10^{16}$ | $9.6 \times 10^{15}$ | $1.9 \times 10^{16}$ | $2.7 \times 10^{16}$ | $3.1 \times 10^{16}$ |
| Ratio | 0.5 | 0.4 | 0.7 | 1.0 | 1.1 |
| LEDs (10 000 lux) | | | | | |
| Total effective photons | $6.3 \times 10^{17}$ | $3.5 \times 10^{17}$ | $3.8 \times 10^{17}$ | $5.9 \times 10^{17}$ | $8.4 \times 10^{17}$ |
| Ratio | 1.1 | 0.6 | 0.6 | 1.0 | 1.4 |
| Halogen (10 000 lux) | | | | | |
| Total effective photons | $1.4 \times 10^{17}$ | $2.2 \times 10^{17}$ | $3.4 \times 10^{17}$ | $8.5 \times 10^{17}$ | $6.0 \times 10^{17}$ |
| Ratio | 0.2 | 0.3 | 0.4 | 1.0 | 0.7 |
| Sunlight (90 000 lux) | | | | | |
| Total effective photons | $5.0 \times 10^{18}$ | $4.0 \times 10^{18}$ | $3.1 \times 10^{18}$ | $4.9 \times 10^{18}$ | $8.3 \times 10^{18}$ |
| Ratio | 1.0 | 0.8 | 0.6 | 1.0 | 1.7 |

LEDs = light-emitting diodes; LWS = long wavelength sensitive; MWS = medium-wavelength sensitive; SWS = short-wavelength sensitive; VS = violet sensitive.
The peak sensitivity (λ max) of each opsin is displayed in column titles. Ratios are normalized to an LWS of 1.
Due to limited data on rod photoreceptor transmittance in the chicken, rod total photons are calculated using only ocular transmittance, relative opsin sensitivity, and aperture corrections.

As shown in Table 3, sunlight and LED light induced a similar photoreceptor activation pattern (as assessed by measuring the total area under the curve [total effective photons across all wavelengths for each photoreceptor family]). In contrast, both halogen and fluorescent lights showed a pattern distinctly different from that of sunlight, with a strong bias toward activation of long-wavelength-sensitive cones.

### Data and Statistical Analysis

As previously described,[42] a power calculation was performed to assess group sizes required to achieve 80% power in observing a 1 D change in refraction when the standard deviation is approximately 0.5 D, which indicated minimum group sizes of 4 animals.

Refractive and axial length measurements are presented as means ± standard error of the means, with the data representing the difference between the treated and the contralateral control eye after treatment. Liquid chromatography tandem mass spectrometry data are presented as the peak area ratio ± standard error of the means.

Before statistical analysis, all data were first tested for normality and homogeneity of variance (Shapiro−Wilk test). All statistical analyses for the effect of treatment were undertaken using a 1-way univariate analysis of variance followed by post hoc Student *t* test with Bonferroni correction for multiple testing. The reproducibility of light protection was analyzed by a Levene test for equality of variances. All statistical analyses were undertaken using the statistical software program SPSS (IBM) with a cutoff of 0.05 for statistical significance.

## Results

### Experiment 1: The Effect of Bright Light on the Rate of Compensation to Negative Lens Wear

When chicks were reared under 500 lux, negative lens wear induced significant myopic shifts in refraction ($F$ [1,179] = 21 605.21, $P < 0.05$) and axial elongation ($F$ [1,35] = 428.02, $P < 0.05$) over the experimental period (Fig 4, Table 4). Light intensity had a significant effect on refractive development in negative lens-treated eyes over the experimental period, showing a slower rate of change with increasing light intensities ($F$ [2, 268] = 1011.664, $P < 0.05$; Fig 4A; Table 4). Specifically, chickens reared under normal indoor lighting levels (500 lux) showed an average rate of change in refraction of −1.48 D per day in response to −10 D lenses, leading to full compensation to the imposed defocus by day 6. In contrast, daily exposure to 20 000 lux for a period of 6 hours per day reduced the average rate of change in refraction to −0.81 D per day ($F$ [1,179] = 526.429, $P < 0.05$), with full compensation delayed by 4 days. This effect was further enhanced by daily exposure to 40 000 lux, which reduced the average rate of change in refraction to −0.54 D per day ($F$ [1,179] = 2146.852, $P < 0.05$), with full compensation delayed by 8 days.

In concordance with the refractive changes observed, illumination levels had a significant effect on axial elongation in negative lens-treated eyes ($F$ [2, 52] = 40.211, $P < 0.05$; Fig 4B; Table 4). Specifically, by the midpoint measure (day 8), chicks kept under 500 lux, in which full refractive compensation had been achieved, showed a change in axial length of +0.82 mm relative to baseline measures ($P < 0.05$). In contrast, at the same timepoint, chicks exposed to 20 000 lux showed a reduced change in axial length of only +0.60 mm ($F$ [1, 35] = 81.961, $P < 0.05$), while those chicks kept under 40 000 lux showed the smallest change (+0.39 mm; $F$ [1, 35] = 13.745, $P < 0.05$).

### Experiment 2: Bright Light Consistently Inhibited the Development of LIM across Batches of Animals

To investigate how consistent the effects of bright light were against the development of LIM, the experiment was repeated across a total of 4 batches of animals. The results demonstrate a significant level of variability in the response seen to 40 000 lux (average rate of refraction per day was −0.55 D, −0.59 D, −0.92 D, and −0.76 D for the 4 experimental repeats, respectively; Fig 5). Although the rate of compensation to negative lens wear was significantly retarded in all experimental repeats relative to that seen under 500 lux, the degree to which this was observed was significantly different between each run for refraction ($F$ [1, 125] = 86.7969, $P < 0.05$; Fig 5; Table S5, available at www.ophthalmologyscience.org) and axial length ($F$ [1, 41] = 38.277, $P < 0.05$; Fig 5; Table S5). In contrast, chicks raised under 500 lux with −10 D lenses did not show significant variation between repeats for refraction ($F$ [1, 125] = 0.055, $P = 0.815$; Fig 5; Table S5). However, these animals did show significant variation in axial length ($F$ [1, 41] = 32.019, $P < 0.05$; Fig S6, available at www.ophthalmologyscience.org; Table S5), although to a lower degree than that seen for high light.

### Experiment 3: The Role of DA on the Effect of Bright Light during Negative Lens Compensation

The protective effect of bright light on the rate of lens compensation was significantly inhibited (refraction 72%, axial growth 77%) by the DA D2-like receptor antagonist spiperone (Fig 7; Table 6). Specifically, the change in refraction and axial length was similar between chicks exposed to 40 000 lux but who received a daily dose of spiperone and those animals reared under 500 lux (refraction: $P = 0.269$, axial length: $P = 0.616$). In contrast, the rate of lens compensation was significantly reduced in chicks exposed to 40 000 lux (refraction: $P < 0.05$, axial length: $P < 0.05$). There was a strong correlation between the changes observed in refraction and axial length ($y = -15.65x$, $R^2 = 0.95$; Fig S8, available at www.ophthalmologyscience.org).

The protection associated with 40 000 lux was unaffected by intravitreal administration of a vehicle solution (refraction: $P = 0.786$, axial length: $P = 0.824$). Similarly, for refraction, administration of the D1-like receptor antagonist SCH-23390 did not attenuate the protective effects of bright light exposure ($P = 0.769$). However, for axial length, although the biological trend is in the correct direction, due to the spread of data, SCH-23390-treated animals were not

Table 4. Changes in Ocular Parameters in Response to LIM (−10 D) at Different Light Intensities (Experiment 1)

| Treatment | Ocular Measure | Days | | | | | |
|---|---|---|---|---|---|---|---|
| | | 0 | 1 | 2 | 3 | 4 | 5 |
| Untreated | Refraction | 3.35 ± 0.04 | 3.45 ± 0.07 | 3.41 ± 0.06 | 3.41 ± 0.07 | 3.27 ± 0.10 | 3.18 ± 0.04 |
| | Axial length | 8.36 ± 0.01 | - | - | - | - | - |
| LIM (500 lux) | Refraction | 3.35 ± 0.04 | 1.58 ± 0.16 | 0.42 ± 0.16 | −1.04 ± 0.23 | −2.41 ± 0.28 | −4.22 ± 0.23 |
| | Axial length | 8.36 ± 0.02 | - | - | - | - | - |
| LIM (20 000 lux) | Refraction | 3.35 ± 0.06 | 2.69 ± 0.16 | 2.04 ± 0.07 | 1.11 ± 0.05 | 0.22 ± 0.13 | −0.70 ± 0.24 |
| | Axial length | 8.34 ± 0.02 | - | - | - | - | - |
| LIM (40 000 lux) | Refraction | 3.25 ± 0.05 | 2.99 ± 0.07 | 2.56 ± 0.13 | 1.65 ± 0.21 | 1.19 ± 0.20 | 0.52 ± 0.17 |
| | Axial length | 8.34 ± 0.02 | - | - | - | - | - |

D = diopters; LIM = lens-induced myopia.
Refraction (D); axial length (mm).
Results are presented as the mean refraction (D) or axial length (millimeters) ± standard error of the mean (n = 6 per condition, per timepoint).

statistically different from either the 40 000 lux ($P = 0.482$) or 500 lux groups ($P = 0.140$).

### Experiment 4: Measures of Ocular DA Levels and the Role of ON-Bipolar Cell Input

After 30 minutes of light exposure in dark-adapted animals, an intensity-dependent increase in vitreal DOPAC levels was observed in otherwise untreated chicks ($F$ [5, 31] = 16.319, $P < 0.05$; Fig 9A, B, Table S7, available at www.ophthalmologyscience.org). Exposure to light, at any intensity, was associated with a significant increase in DOPAC levels above those seen in chicks under 0 lux ($P < 0.05$). A further increase in DOPAC levels was observed in chicks exposed to 10 000 to 40 000 lux compared with those under 50 lux ($P < 0.05$), and chicks under 20 000 and 40 000 lux compared with those under 500 lux ($P < 0.05$). The relationship between light intensity and DOPAC levels is best explained by a logarithmic function ($y = 8 \times 10^{-5} \log x + 0.0002$, $R^2 = 0.70$). When the x-axis is logged (Fig 9B), it is best explained by a linear function ($y = 2.31 \times 10^{-4}x - 5.63 \times 10^{-5}$, $R^2 = 0.70$).

To investigate the retinal pathways underlying light's modulation of dopaminergic amacrine cells, the drug AP4 was administered to remove subsets of bipolar cell function (namely, rod and ON-bipolar cell pathways). When single-flash ERG recordings were undertaken, AP4 induced a significant reduction in B-wave amplitude at all light intensities tested ($F$ [1, 9] = 34.6310, $P < 0.05$; Fig 10). Further investigation via flicker ERG analysis demonstrated that, as expected, this reduction in B-wave amplitude could be attributed to a significant disruption of rod-bipolar cell function (0.5 Hz, $P < 0.05$) and cone ON-bipolar cell function (5−10 Hz, $P < 0.05$), but not cone OFF-bipolar cell function (15−30 Hz, $P = 0.363$). Due to the significant reduction observed in B-wave amplitude in AP4-treated animals, oscillatory potentials were unable to be extracted. Administration of a vehicle solution alone in a separate experiment did not significantly affect any of the ERG responses measured ($F$ [1, 9] = 0.965, $P = 0.509$; Fig S11, available at www.ophthalmologyscience.org).

After 2 hours, the fitment of diffusers or negative lenses under 500 lux was associated with a significant (FDM: 40%, LIM: 48%) reduction in vitreal DOPAC levels relative to contralateral control values ($P < 0.05$; Fig 12, Table 8). This downregulation was blocked by exposure to 40 000 lux, with treated eyes not significantly different from contralateral controls (FDM: $P = 0.123$, 13% reduction; LIM: $P = 0.409$, 18% reduction). In contralateral control eyes exposed to bright light relative to values seen in otherwise untreated animals under 500 lux ($P < 0.05$), DOPAC levels were also observed to rise. In contrast, the increase in vitreal DOPAC associated with 40 000 lux in FDM-treated and LIM-treated eyes was lost with the administration of AP4 ($P < 0.05$, FDM: 37% reduction from contralateral control values, LIM: 37% reduction from contralateral control values).

## Discussion

As reported for FDM, daily exposure to bright light inhibited the development of LIM in an intensity-dependent manner. This occurred, at least in part, through a dopaminergic mechanism. However, unlike FDM, in which myopic growth can be halted, bright light only delayed compensation to negative lenses (this concurs with previous findings[19]), the implications of which are discussed here.

### The Dopaminergic System Underlies the Protective Effects of Bright Light

The dopaminergic system plays a key role in the development of both forms of experimental myopia,[43] and for FDM, the protective effects of bright light are driven by the intensity-dependent release of this catecholamine.[19] In this study, we demonstrate that bright light also inhibits the development of LIM, at least in part, through a similar dopaminergic mechanism. A direct link between DA release and the antimyopic effect of light has not been

Table 4. (Continued.)

| Days | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| *6* | *7* | *8* | *9* | *10* | *11* | *12* | *13* | *14* |
| 3.21 ± 0.08 | 3.12 ± 0.07 | 3.14 ± 0.05 | 3.11 ± 0.05 | 3.02 ± 0.05 | 3.00 ± 0.03 | 2.99 ± 0.05 | 2.96 ± 0.04 | 2.95 ± 0.04 |
| - | - | 8.61 ± 0.02 | - | - | - | - | - | 8.78 ± 0.01 |
| −5.55 ± 0.12 | −5.70 ± 0.09 | −5.61 ± 0.10 | −5.65 ± 0.11 | −5.67 ± 0.10 | −5.84 ± 0.13 | −5.74 ± 0.15 | −6.00 ± 0.09 | −6.16 ± 0.08 |
| - | - | 9.18 ± 0.03 | - | - | - | - | - | 9.26 ± 0.02 |
| −1.52 ± 0.25 | −2.30 ± 0.26 | −3.15 ± 0.27 | −4.42 ± 0.22 | −5.53 ± 0.14 | −5.52 ± 0.13 | −5.87 ± 0.19 | −6.02 ± 0.12 | −6.12 ± 0.10 |
| - | - | 8.94 ± 0.02 | - | - | - | - | - | 9.30 ± 0.02 |
| 0.04 ± 0.09 | −0.28 ± 0.17 | −0.69 ± 0.13 | −1.32 ± 0.16 | −2.04 ± 0.19 | −2.93 ± 0.23 | −3.93 ± 0.23 | −4.84 ± 0.20 | −5.84 ± 0.17 |
| - | - | 8.73 ± 0.01 | - | - | - | - | - | 9.25 ± 0.01 |

assessed outside of the chicken. One would presume such a mechanism is conserved, as DA release shows an intensity-dependent response in all species studied.[44,45] It is interesting to note that bright light had a similar effect on DA levels in FDM and LIM, yet physiologically, light shows a stronger protective effect on the axial and refractive change associated with diffuser wear[14] relative to negative lenses. This suggests that DA levels alone are not the only driver of growth outcomes.

The ability of the drug AP4 to block light-induced DA release (in either model of experimental myopia) suggests that protection is driven through 1 of 2 bipolar cell pathways

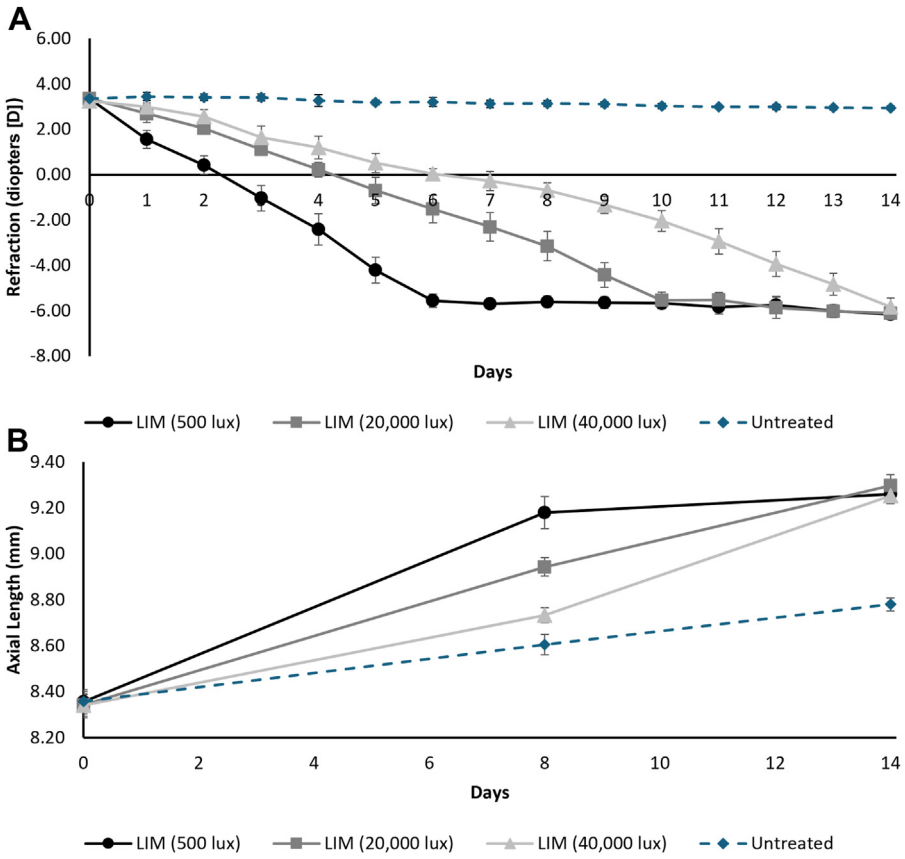



**Figure 4.** The effect of bright light on refraction and axial length in response to LIM (−10 D) (experiment 1). (**A**) The rate of compensation to negative lens wear was significantly delayed the greater the light intensity to which animals were exposed (20 000 lux and 40 000 lux). (**B**) As with refraction, bright light inhibited the rate of axial elongation associated with LIM. LIM = lens-induced myopia.

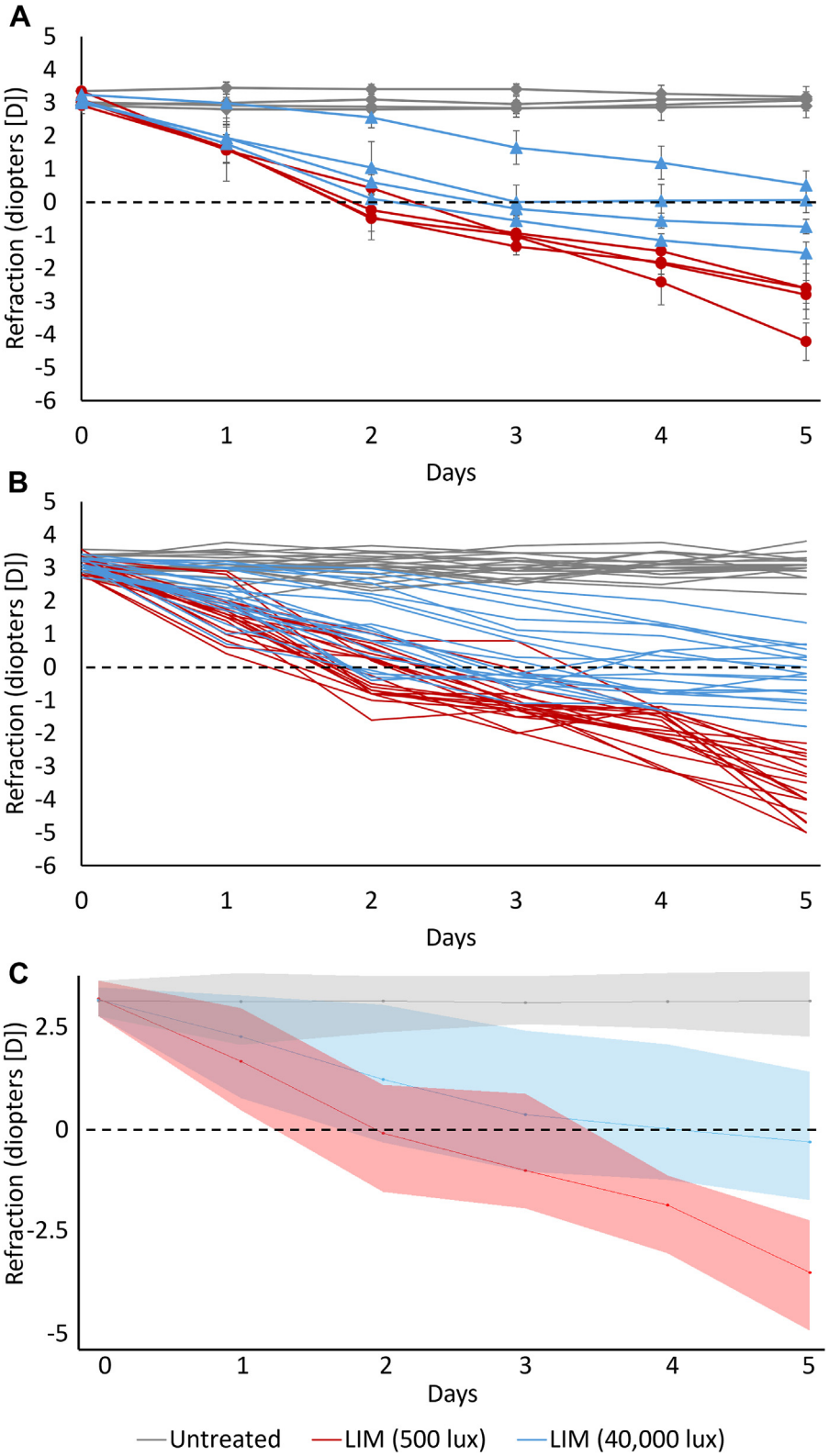



**Figure 5.** The reproducibility of the antimyopic effects of bright light in negative lens-wearing animals (experiment 2). Although variability was seen between the 4 batches tested, bright light significantly inhibited the rate of compensation to negative lenses in each experiment. The variability in refraction is depicted in 3 ways: (**A**) the average refraction for each batch within each condition; (**B**) the refraction for individual animals within each condition; and (**C**) the range in refraction, with the colored lines indicating the average refraction across the 4 experimental batches. LIM = lens-induced myopia.

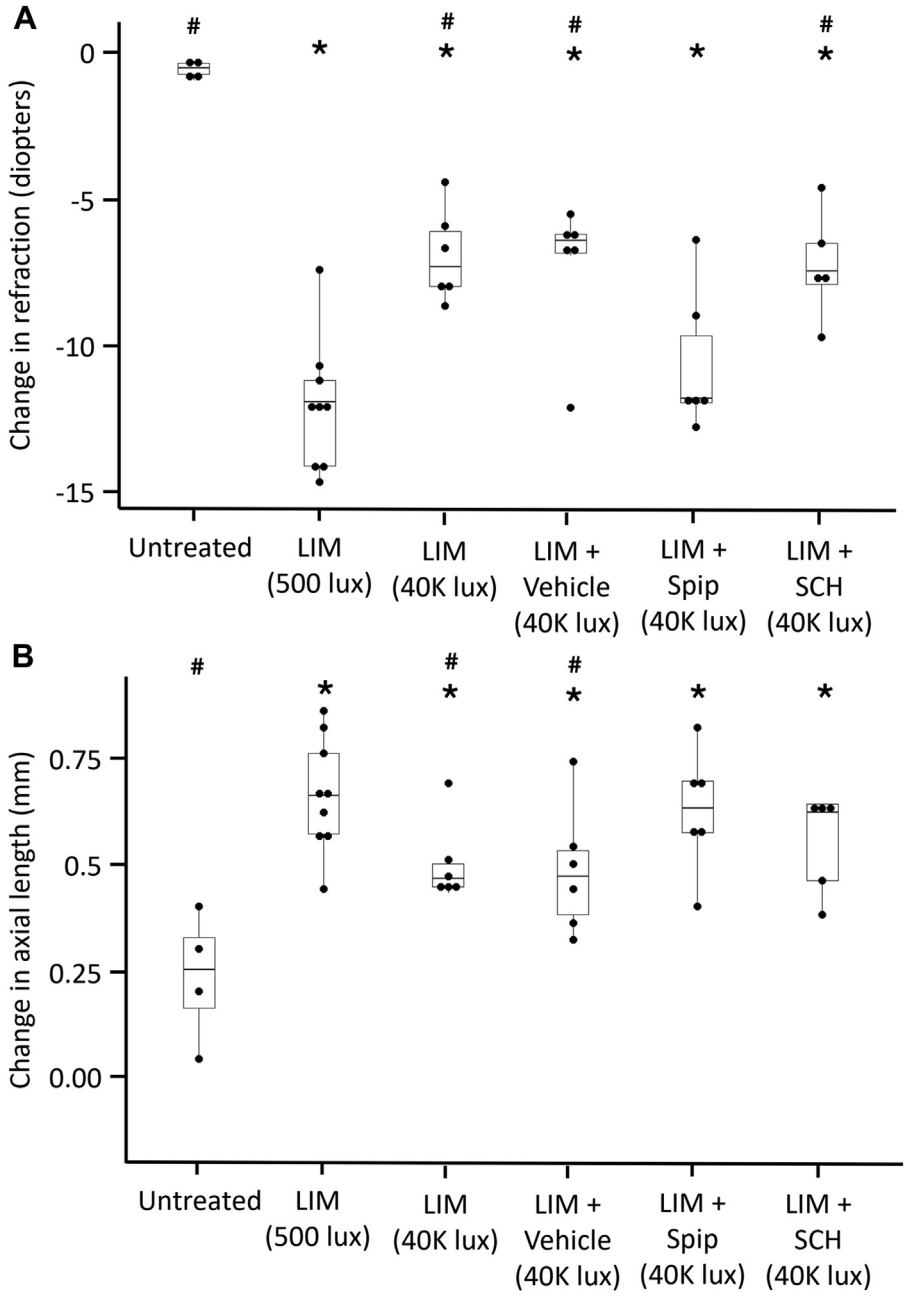



**Figure 7.** The effects of dopamine receptor blockade on the antimyopic effects of bright light (experiment 3). Changes in (**A**) refraction and (**B**) axial length from the commencement to end-of-treatment measures in chicks undergoing no treatment (untreated), LIM under 500 lux, LIM under 40 000 lux (abbreviated as 40K lux), and LIM under 40K lux and injected daily with either the D2-like dopamine receptor antagonist spiperone (Spip), the D1-like dopamine receptor antagonist SCH-23390 (SCH), or a vehicle solution. Box plots represent the interquartile ranges of each group, while black dots represent individual values. For the y-axes, change in refraction and axial length represent the final treatment value for each animal minus its baseline. Statistics represent a significant difference to *untreated chicks, #LIM only animals reared under 500 lux ($P < 0.05$). LIM = lens-induced myopia.

previously implicated in myopia development[46–52]: that of rod bipolar cells or ON-bipolar cells. Unfortunately, we are unable (from this data alone) to further distinguish which of these pathways is critical. Further, AP4 will inhibit bipolar cell communication to much of the retinal circuitry, not just DA amacrine cells. Therefore, we cannot confirm whether the changes observed in DA cell activity are due to direct input from bipolar cells or due to other alterations in retinal cell activity. The current results also cannot discount a role for intrinsically photosensitive retinal ganglion cells (ipRGCs). While AP4 does not affect the activation of ipRGCs through their intrinsically photosensitive properties, it can attenuate their activation via synaptic input from classical photoreceptor pathways.[53] Therefore, if AP4's

Table 6. The Effect of a D1 (SCH-23390) and D2 (Spiperone) Dopamine Receptor Antagonist on the Protective Effects of Bright Light against LIM (Experiment 3)

| Condition | Refraction (Diopters) | | | Axial Length (mm) | | |
|---|---|---|---|---|---|---|
| | *Day 0* | *Day 5* | *Change* | *Day 0* | *Day 5* | *Change* |
| Untreated | 3.46 ± 0.09 | 2.90 ± 0.10 | −0.56 ± 0.14† | 8.39 ± 0.04 | 8.62 ± 0.03 | 0.23 ± 0.08† |
| LIM (500 lux) | 3.10 ± 0.13 | −8.91 ± 0.72 | −12.02 ± 0.71* | 8.42 ± 0.03 | 9.08 ± 0.04 | 0.66 ± 0.05* |
| LIM (40 000 lux) | 3.26 ± 0.17 | −3.65 ± 0.70 | −6.91 ± 0.67*,† | 8.40 ± 0.03 | 8.90 ± 0.04 | 0.50 ± 0.01*,† |
| LIM + Vehicle (40 000 lux) | 3.20 ± 0.11 | −4.05 ± 0.86 | −7.24 ± 0.14*,† | 8.40 ± 0.03 | 8.89 ± 0.05 | 0.48 ± 0.04*,† |
| LIM + Spiperone (40 000 lux) | 3.28 ± 0.08 | −7.33 ± 0.72 | −10.61 ± 0.67* | 8.39 ± 0.02 | 9.01 ± 0.03 | 0.63 ± 0.05* |
| LIM + SCH-23390 (40 000 lux) | 3.37 ± 0.07 | −3.86 ± 0.77 | −7.23 ± 0.84*,† | 8.38 ± 0.04 | 8.93 ± 0.06 | 0.55 ± 0.05* |

LIM = lens-induced myopia.
Vehicle: 0.1% ascorbic acid, change: day 5 measures − day 0 measures. Data are presented as the means ± standard error. Statistics ($P < 0.05$) represent a significant difference to *untreated chicks, †LIM only animals reared under 500 lux.

effects on DA observed in this study are elicited via ipRGCs, this would still be mediated by rod and ON-bipolar cells. It should also be noted that, given the role of ipRGCs in diurnal DA fluctuations, this analysis may have captured only the immediate effects of bright light on DA levels. This leaves open the possibility that a melanopsin-dependent mechanism contributes to any sustained dopaminergic response to light.

Other than dopaminergic cell activity, there are several other factors associated with bright light exposure that could explain its antimyopic effects, including changes in focal depth, physical activity level of the animals, increased optic flow, changes in spatial and temporal image structure, as well as the activation of retinal cell types other than dopaminergic cells. These factors were not investigated presently, although previous studies have shown that a number of these parameters are unlikely to drive the protective effects of light (i.e., pupil size, activity level, and optic flow[13]).

## Differences between FDM and LIM

Although the 2 forms of experimental myopia share many biochemical and physiological similarities (for review, see[43,54]) and are likely regulated by the same retinal pathways,[54–56] there are also clear differences between these 2 paradigms,[54–56] as was again evident in this study.

A hypothesis to explain the difference in the responses of FDM and LIM is that there are 2 control systems in operation. The first mechanism is the light intensity-dependent and contrast-dependent regulation of DA release. In this mechanism, a loss of contrast caused by the fitment of a lens or diffuser will lead to a reduction in DA levels (as observed experimentally[10,57]). In FDM, DA release may be further downregulated by the reduction in light transmittance observed (∼15%). Together, this reduction in DA (in both LIM and FDM) acts as a growth enhancement signal, but with no defined growth end point. The second mechanism is driven by the detection of a shifted focal plane (due to a change in optical power). This DA-independent defocus cue has a defined end point (compensation). In FDM, there is no focal cue; thus, the eye is without a defined growth end point (open-loop). In this state, the system appears to be more sensitive to modulation of the first mechanism (dopaminergic activity). Thus, when DA release is stimulated by bright light exposure,[58] this growth inhibitory cue is able to abolish the development of FDM.[14] In contrast, while light-induced DA release will also act as a growth inhibitory signal in LIM, it is not able to override the second (and stronger) focal cue that drives the eye toward compensation (closed-loop).[19]

A similar hypothesis can be applied to the growth response observed in lens-induced hyperopia. Upon fitment of positive lenses, a loss of contrast will cause a reduction in DA levels (as has been observed experimentally[59]). On its own, this reduction in DA levels would act as a growth enhancement signal. However, due to the presence of a more potent, defocus-dependent mechanism, ocular growth rates slow to compensate for the myopic defocus. When DA release is stimulated by bright light exposure, this will act as an additional growth inhibitory signal (as seen for FDM and LIM). This is what is observed experimentally, with bright light enhancing the rate of compensation to plus lenses.[19]

The differences between FDM and LIM (as well as lens-induced hyperopia) again raise a familiar question (to which there is no clear answer): which form of experimental myopia, if any, best represents the human condition? It has recently been postulated by Schaeffel and Swiatczak[60] that normal ocular development acts like a closed-loop system in which there is a target refractive outcome (between +0.5 and +1.5 D). In contrast, the inability of the myopic retina to inhibit its own growth in response to hyperopic defocus and its reduced sensitivity to several visual manipulations are more akin to an open-loop system. An interesting point raised by Schaeffel and Swiatczak[60] is that understanding human myopia may be better served by studying the growth inhibitory pathways associated with lens-induced hyperopia rather than modeling the pathways associated with growth enhancement (as seen in FDM and LIM).

## Optimal Lighting Parameters

There is still much we do not understand about the optimal lighting parameters required to effectively inhibit myopic growth (i.e., timing, duration, spectral composition, and temporal patterning) or how light may interact with other

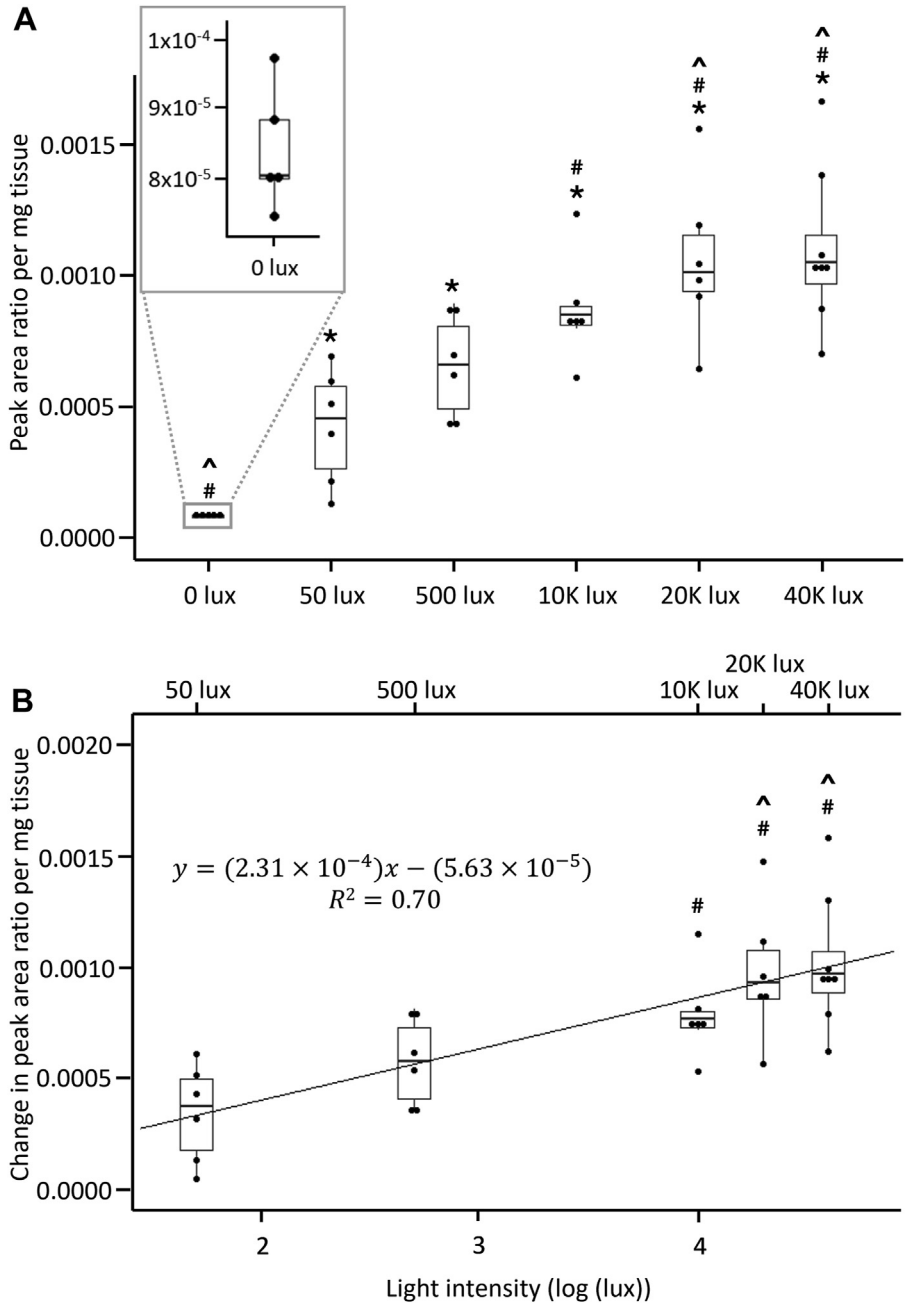



**Figure 9.** Dose-dependency between light intensity and vitreal DOPAC levels (experiment 4). The DOPAC was measured 30 minutes after light exposure (50 lux, 500 lux, 10 000 lux [10K lux], 20 000 lux [20K lux], and 40 000 lux [40K lux]) in otherwise untreated animals after overnight dark adaptation. Data are plotted in (**A**) ordinal and (**B**) logarithmic scales. Box plots represent the interquartile ranges of DOPAC peak area ratios (DOPAC: internal standard) per mg of tissue; black dots represent individual values. Data in panel B are plotted as the change in peak area ratio from darkness (0 lux). Statistics represent a significant difference to chicks exposed to *0 lux, #50 lux, and ^500 lux ($P < 0.05$). DOPAC = 3,4-dihydroxyphenylacetic acid.

environmental/visual cues postulated to be critical for growth regulation.

Currently, animal data would suggest that ≥2 hours of continuous bright light exposure each day is required to produce a meaningful antimyopic effect, with no greater efficacy seen after 6 hours of exposure.[61] There also appears to be a temporal component to this protection, with greater efficacy observed when light is given in bursts rather than 1 continuous period (for review, see[43]). For example, Lan et al[62] have reported that exposure to intermittent bright light (15 000 lux) with a frequency cycle of 1:1 or 7:7 minutes was significantly more effective at inhibiting FDM compared with continuous bright light. Consistent with our current findings, this may be associated with the

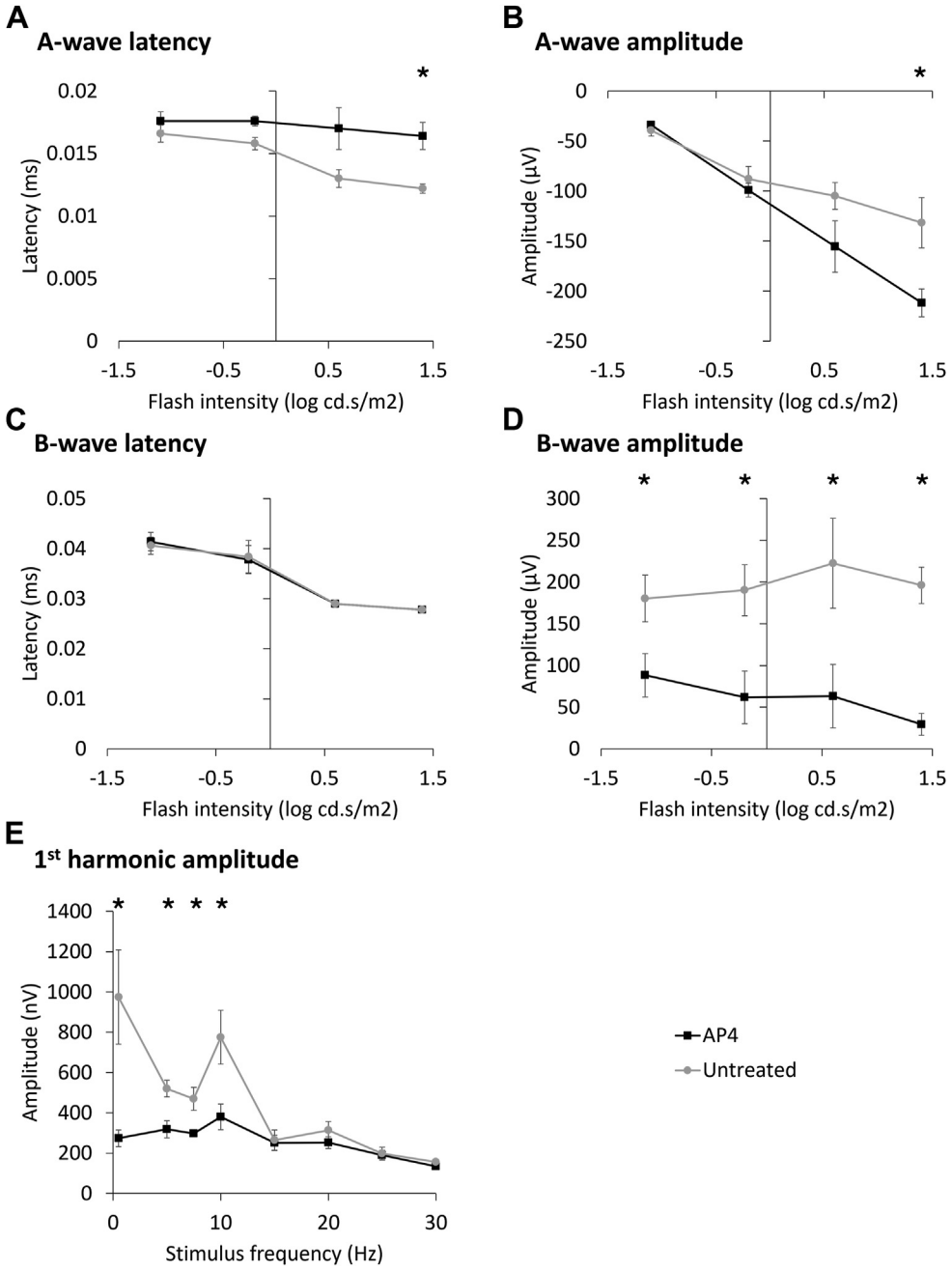



**Figure 10.** Effects of AP4 administration on electroretinogram responses (experiment 4). The (**A**) A-wave latency, (**B**) A-wave amplitude, (**C**) B-wave latency, (**D**) B-wave amplitude, and (**E**) first harmonic amplitude were measured in chicks after intravitreal injections of AP4. Data are presented as the means ± standard error. Statistics represent a significant difference to otherwise untreated animals (*$P < 0.05$). AP4 = L-(+)-2-amino-4-phosphonobutyric acid.

greater DA release induced by stimulation of ON pathways at higher temporal frequencies. However, it is important to note that although the total duration of bright light exposure is the same when given intermittently or continuously, the intermittent exposure profile will cover a larger portion of the day. This is important, as several studies have observed that treatment efficacy varies with time of day, being more effective when light is presented in the middle of the day[63] or evening[64] when compared with the morning.

One area of significant interest is how the spectral composition of "white light" might alter its efficacy. Monochromatic experiments,[43] as well as experiments involving the spectral composition of artificial indoor "white" light being altered,[65–69] have demonstrated that modulating the wavelength in which animals are reared can

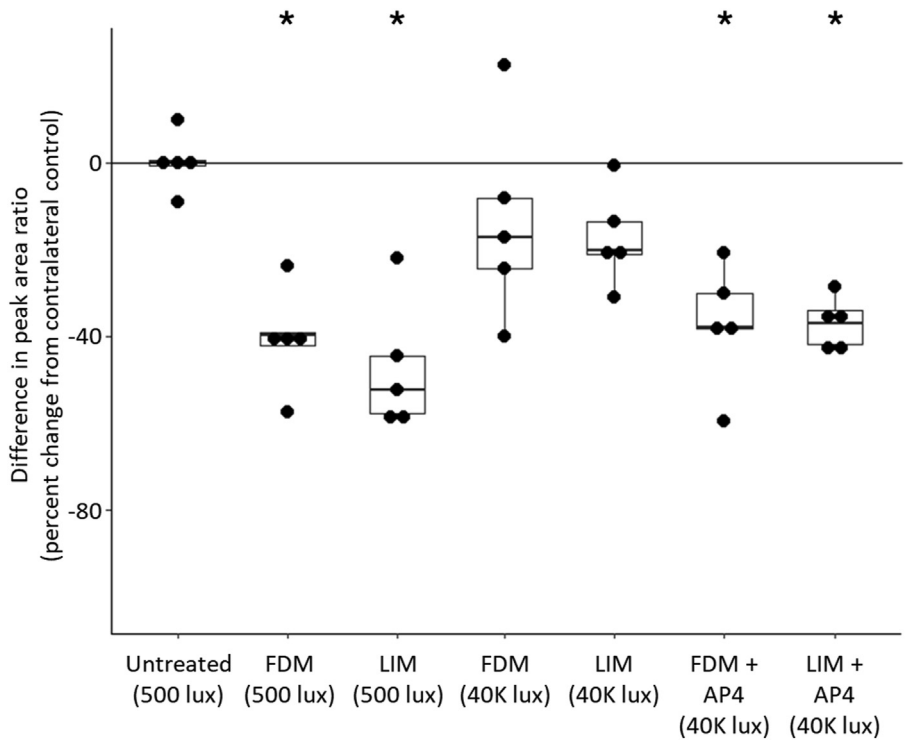



**Figure 12.** Effects of AP4 treatment and bright light on vitreal DOPAC levels in animals developing FDM and LIM (experiment 4). The DOPAC was measured 2 hours after the commencement of treatment. Box plots represent the interquartile ranges of the percent change in DOPAC peak area ratio (DOPAC: internal standard) between treated and contralateral control eyes. Black dots represent individual values. Statistics represent a significant difference between treated and contralateral control eyes (*$P < 0.05$). AP4 = L-(+)-2-amino-4-phosphonobutyric acid; DOPAC = 3,4-dihydroxyphenylacetic acid; FDM = form-deprivation myopia; LIM = lens-induced myopia.

inhibit or enhance myopic growth (although there are differences in distinct species). However, as discussed later, what role spectral composition plays in the antimyopic effects of "bright" light is unclear. The current literature is unable to address this question due to significant differences in experimental design between studies (e.g., lighting source/parameters, mode of myopia induction, treatment period, species investigated, etc.). It should also be noted that the majority of lighting systems used in animal studies do not emit outside of the visible spectrum, thus providing limited information on the role of UV or IR in growth inhibition.

Another important question is how light interacts with other visual and environmental cues that appear critical for growth regulation. For example, could an additive effect be seen when concurrently exposing an individual to multifocal peripheral defocus, which has shown strong treatment efficacy,[70] and bright light? In chicks, exposure to bright light enhances the growth inhibition associated with full-field myopic defocus (plus lenses), suggesting

Table 8. Effects of AP4 Treatment and Bright Light on Vitreal DOPAC Levels in Treated and Control Eyes (Experiment 4)

| | DOPAC PAR | | |
|---|---|---|---|
| **Condition** | *Treated Eye* | *Contralateral Control* | *Difference* |
| Untreated (500 lux) | 0.099 ± 0.004$^{\dagger}$ | 0.099 ± 0.005$^{\dagger}$ | 0.000 ± 0.003 |
| FDM (500 lux) | 0.054 ± 0.009$^{*,\ddagger}$ | 0.088 ± 0.008 | −0.034 ± 0.003$^{\ddagger}$ |
| LIM (500 lux) | 0.068 ± 0.010$^{*,\ddagger}$ | 0.130 ± 0.018 | −0.062 ± 0.014$^{\ddagger}$ |
| FDM (40 000 lux) | 0.099 ± 0.008$^{\dagger}$ | 0.116 ± 0.006$^{\dagger,\ddagger}$ | −0.018 ± 0.013 |
| LIM (40 000 lux) | 0.141 ± 0.022$^{\dagger}$ | 0.172 ± 0.028$^{\dagger,\ddagger}$ | −0.031 ± 0.013 |
| FDM + AP4 (40 000 lux) | 0.076 ± 0.004* | 0.124 ± 0.010$^{\dagger}$ | −0.049 ± 0.013$^{\ddagger}$ |
| LIM + AP4 (40 000 lux) | 0.121 ± 0.015* | 0.193 ± 0.026$^{\dagger,\ddagger}$ | −0.072 ± 0.013$^{\ddagger}$ |

AP4 = L-(+)-2-amino-4-phosphonobutyric acid; DOPAC = 3,4-dihydroxyphenylacetic acid; FDM = form-deprivation myopia; LIM = lens-induced myopia; PAR = peak area ratio (DOPAC:internal standard).
The difference represents the treated eye PAR minus that of the contralateral control. Data are presented as the means ± standard error. Statistics represent a significant difference to *contralateral control, $^{\dagger}$FDM or LIM (500 lux) values, and $^{\ddagger}$500 lux control values ($P < 0.05$).

the potential for such an additive effect.[19] There are also several other parameters, not associated with light, believed to be critical to the antimyopic effects of being outdoors. Could an additive effect be seen when combined with bright light? For example, Flitcroft[71] has postulated that myopic growth may be due to alterations in the spatial makeup of indoor scenes when compared with outdoor environments. To this end, a recent study by Lan et al[72] demonstrated that increasing the spatial information in classrooms (to match that of an outdoor scene) helped maintain the hyperopic reserve of children over a 12-month period. This raises the question of whether an interaction between light and spectral makeup could be advantageous for myopia management. We know that spatial and contrast sensitivity is enhanced with increasing illumination (although a ceiling is reached at daylight illuminance levels [10 000 lux][73]). Moreover, the enhancement of the contrast sensitivity function over increasing illuminances is driven by higher dopaminergic stimulation within the retina.[74] At scotopic intensities, this increase is seen across all spatial frequencies, while at photopic intensities, it is primarily seen at higher spatial frequencies.[75] Thus, there is a potential for an interaction effect between bright light and spatial frequency makeup.

### Differences in the Spectral Composition of White Light

Given the increasing evidence that spectral cues play a critical role in ocular development, we looked closely at the spectral output, when compared with sunlight, of different forms of artificial "white light." Sunlight is considered wide-band (emitting throughout the visible spectrum and into UV and IR), with its spectral distribution affected by the solar elevation angle and atmospheric composition.[76] In contrast, artificial lighting systems usually have a narrower spectral distribution.[77] For example, fluorescent lights have a series of narrow band peaks at specific wavelengths, the white LEDs emit a broader, 2-peak spectra, while halogen lighting produces a smoother spectral distribution (but one that is heavily biased toward longer wavelengths). With regard to what this spectral distribution means for total effective photons available to chicken photoreceptors, a similar activation pattern was observed between (early afternoon) sunlight and the bright light LEDs used in this study (with a near-equal activation of all opsin types analyzed). In contrast, fluorescent and halogen lights elicited a very different photoreceptor activation pattern, biasing heavily toward the red end of the spectrum. This raises the question of whether the protection afforded by artificial bright "white light" is purely intensity-dependent or whether spectral composition also plays a role. Previous studies (predominantly in chicks) have demonstrated that differences in the spectral composition of artificial indoor white light (100–1000 lux) can affect normal[66,68] and myopic[65,67,69] development, with a trend toward greater inhibition of growth in blue-biased light. To the best of our knowledge, the role spectral composition plays in the protection afforded by "bright" (>5000 lux) white light has not been heavily investigated. However, if we compare the protection afforded in chicks by 15 000 lux of "white light" generated by LEDs[14] with that generated by a halogen system,[13,19] although the spectral composition is quite different between the 2 lighting systems, the antimyopic effect is extremely similar (LED, ~63% protection [refraction] vs. halogen, ~65% protection [refraction]). This would suggest that total illuminance, rather than the spectral makeup of white light, is responsible for this protection. However, future studies will need to investigate further the role of spectral makeup in any ocular growth effects associated with bright "white light."

### Clinical Translation

Although light exposure has been strongly correlated with the antimyopic effects of time outdoors in patient populations, we currently lack direct evidence of the role luminance levels play in humans. To be able to directly test this, studies must be designed to isolate changes in luminance from the other confounding factors associated with time outdoors noted above. One such study design would be to increase the lighting levels in selected classrooms to between 5000 and 10 000 lux. With the increasing use of LED technology, this is now more feasible, overcoming historical issues related to heat load and electricity costs. Although such a study would still have confounding factors, this provides the most practical means of testing the role of light intensity in a school cohort.

This study demonstrates that, like that observed for FDM, bright light inhibits the development of LIM in an intensity-dependent manner. The protective effects of bright light observed in this study were mediated through a dopaminergic mechanism, similar to that seen in FDM. However, bright light was only able to slow the rate of compensation to negative lenses and exhibited a lower degree of protection than that seen in FDM. This would suggest that there are mechanistic differences between the 2 models, most likely associated with the presence of a defocus-driven end point for growth in LIM.

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## Footnotes and Disclosures

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ANIMAL SUBJECTS: Authorization to conduct experiments using animals was approved by the University of Canberra Animal Ethics Committee under the ACT Animal Welfare Act 1992 (project number: CEAE 2098) and conformed to the Association for Research in Vision and Ophthalmology Statement for the Use of Animals in Ophthalmic and Vision Research.

Author Contributions:

Conception and design: Karouta, Thomson, Morgan, Ashby

Data collection: Karouta, Thomson, Morgan, Ashby

Analysis and interpretation: Karouta, Thomson, Morgan, Ashby

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Abbreviations and Acronyms:
**AP4** = L-(+)-2-amino-4-phosphonobutyric acid; **D** = diopters; **DA** = dopamine; **DOPAC** = 3,4-dihydroxyphenylacetic acid; **ERG** = electroretinogram; **FDM** = form-deprivation myopia; **ipRGCs** = intrinsically photosensitive retinal ganglion cells; **IR** = infrared; **LED** = light-emitting diode; **LIM** = lens-induced myopia; **UV** = ultraviolet.

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